Week 8: Integrate Kubernetes with Docker

Aim:

To enable Kubernetes on Docker Desktop and prepare system to run Kubernetes commands.

Kubernetes

Kubernetes, also known as K8s, is an **open-source system for automatingdeployment, scaling, and management of container-based applications** in different kinds of environments like physical, virtual, and cloud-native computing foundations.

Some of the benefits of Kubernetes

 Container Orchestration: Kubernetes manages the placement and scheduling of containers across a cluster of machines, ensuring that the desired state of the application is maintained.

 Scalability: Kubernetes allows applications to scale horizontally by automatically adding or removing containers based on resource utilization and user-defined rules.

 Self-Healing: Kubernetes monitors the health of containers. It automatically restarts failed containers or replaces them with new ones to ensure the desired state of the application is maintained.

 Rolling Updates and Rollbacks: Kubernetes supports rolling updates, allowing applications to be updated with minimal downtime. If any issues arise, it also facilitates rollbacks to previous versions.

 Storage Orchestration: Kubernetes provides a way to manage persistent storage volumes and attach them to containers as needed.

Differences between Docker and Kubernetes

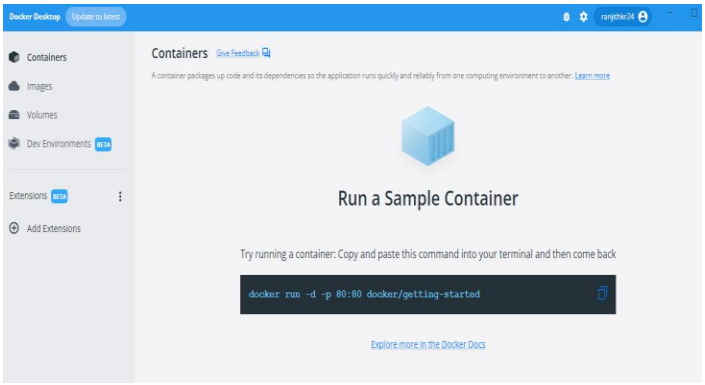
Docker is used for **building, shipping, and running containers**—it provides the means to create isolated environments for applications.

On the other hand, Kubernetes focuses on orchestrating containers, meaning it helps **manage, scale, and ensure the smooth operation of large collections of containers**.

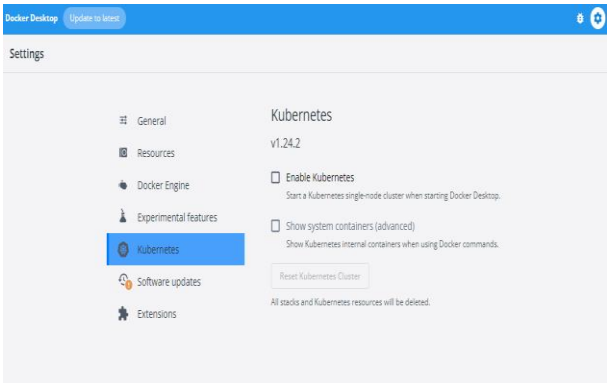
Docker manages individual containers, whereas Kubernetes manages multiple containers across clusters.

Integrating Kubernetes with Docker

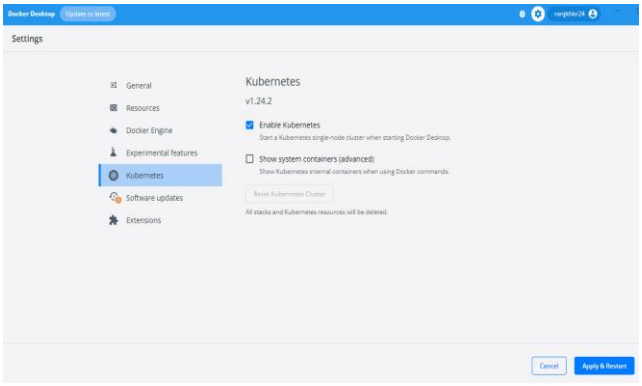
1. **Start the Docker Desktop**



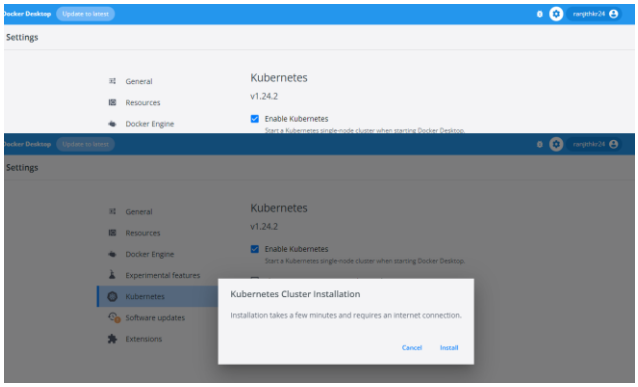
2. **Select Settings on the top right side from the Docker Desktop Dashboard**



3. Select Kubernetes from the left side bar
4. Check the Enable Kubernetes checkbox



5. Select Apply & restart to save the settings
6. Select Install to confirm



7.Go to Kubernetes website ,https://kubernetes.io/releases/download/#binaries

Download **kubectlconvert.exe for Windows OS**

8 Save this file **kubectlconvert.exe**in a folder**and copy the folder into**

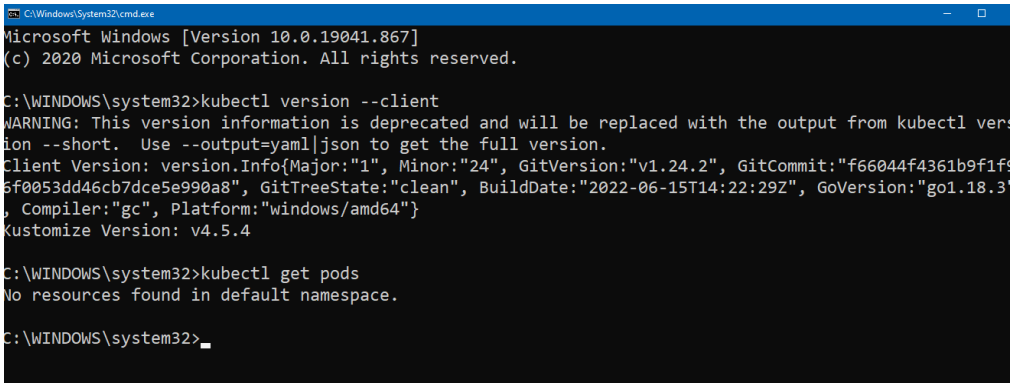
C//Windows//System32

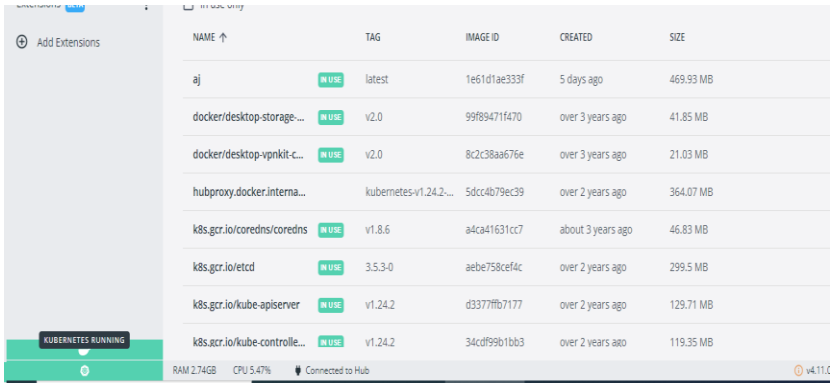
9. Set the path in environment path variable **for the kubectlconvert.exe file**

10. Open command prompt and type the following commands

kubectl version –client

kubectl get pods




Add Extensions
NAME
TAG
IMAGE ID
CREATED
SIZE
aj
IN USE
latest
1e61d1ae333f
5 days ago
469.93 MB
docker/desktop-storage-...
IN USE
v2.0
99f89471f470
over 3 years ago
41.85 MB
docker/desktop-vpnkit-c...
IN USE
v2.0
8c2c38aa676e
over 3 years ago
21.03 MB
hubproxy.docker.interna...
kubernetes-v1.24.2-...
5dcc4b79ec39
over 2 years ago
364.07 MB
k8s.gcr.io/coredns/coredns
IN USE
v1.8.6
a4ca41631cc7
about 3 years ago
46.83 MB
k8s.gcr.io/etcd
IN USE
3.5.3-0
aebe758cef4c
over 2 years ago
299.5 MB
k8s.gcr.io/kube-apiserver
IN USE
v1.24.2
d3377ffb7177
over 2 years ago
129.71 MB
k8s.gcr.io/kube-controlle...
IN USE
v1.24.2
34cdf99b1bb3
over 2 years ago
119.35 MB
KUBERNETES RUNNING
RAM 2.74GB
CPU 5.47%
Connected to Hub